

Geo3: Earthquakes, Volcanoes, Minerals & Rocks



Part 1: Earthquakes

When you have completed studying Part 1 of this eBook, you should:

1. Be able to relate earthquake generation, location, depth, and size to processes at tectonic plate boundaries.
2. Know the types of seismic waves generated in earthquakes, and how earthquake magnitude is measured.

Part 2: Volcanoes



When you have completed studying Part 1 of this eBook, you should be able to answer these questions:

1. How do plate tectonic processes relate to the origin of magmas and volcanic eruptions?
2. What impacts do earthquakes and volcanoes have on humans, and how we can prepare for these hazards?



Part 3: Minerals and Rocks

When you have completed studying Part 1 of this eBook, you should:

1. Know the major chemical composition of each Earth layer, and explain the resulting composition of the mineral and rocks
2. Be able to describe the role of each family rock into the rock cycle.

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- US Geological Survey's webpage, [Animations for Earthquake Terms and Concepts](http://earthquake.usgs.gov/learn/animations/) (<http://earthquake.usgs.gov/learn/animations/>)
- [US Geological Survey FAQ: Measuring Earthquakes](http://www.usgs.gov/faq/taxonomy/term/9828) (<http://www.usgs.gov/faq/taxonomy/term/9828>)
- US Geological Survey: [Volcano Web Shorts](http://volcanoes.usgs.gov/observatories/cvo/web_shorts_videos.html) (http://volcanoes.usgs.gov/observatories/cvo/web_shorts_videos.html)

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G4.1

Earthquakes and Volcanoes: Introduction

A consequence of plate tectonics on planet Earth is the generation of earthquakes and volcanoes. The location and characteristics of these geologic phenomena are explained well in the context of plate tectonic theory – we understand where and why these events occur, and the general size and features of what will happen.

Tectonic processes cause earthquakes and volcanoes over geologic time, but knowing the timing of these events on human timescales of minutes, days, and weeks is difficult. Large volcanic eruptions are often preceded by warning signs, whereas faults provide no indication of a pending rupture that will generate an earthquake.



Figure 12.6b
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In this lesson, we will examine how earthquakes and volcanoes occur, and what measures humans can take to prepare for these inevitable events.



Figure 12.3
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G4.2

Earthquakes

What is an Earthquake?

When rocks under stress suddenly fail along a fault in the Earth, the rupture and movement of rocks releases energy that generates seismic waves – the shaking we feel in an earthquake. Earthquakes occur continuously in the Earth's crust, but most are small and not felt. Large earthquakes occur less frequently but can have devastating effects. Active faults that generate earthquakes are generally, but not always, associated with plate tectonic boundaries.

Watch this video animation [2:00]. It illustrates the relationship between earthquakes and the three different types of plate boundaries. Think about plate motions and the direction of stress build-up along faults. Note the locations and sizes of

EQ_plates_animation



earthquakes
along faults
at different
plate
boundaries.

2

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A quick glance at the distribution of large earthquake epicenters immediately shows that their locations are not randomly distributed over the globe.

They are highly concentrated along linear to arcuate bands, particularly adjacent to the edges of continents and along mid-ocean ridges. These bands mark the edges of the tectonic plates.

Quakes are common at all three types of plate boundaries.

At divergent boundaries, tensional forces break up the lithosphere along normal faults.

This results in abundant shallow-focus earthquakes.

At convergent boundaries, dense oceanic plates descend into the mantle.

This leads to shallow, intermediate, and deep-focus quakes as the plate makes its way downward at an angle.

The deepest focus quakes will therefore have epicenters well into the overriding plate, such as in western South America.

Transform boundaries have numerous shallow to intermediate focus quakes as lithospheric plates slide past one another.

The San Andreas fault is the best known example of this.

Because earthquake activity is so concentrated along plate boundaries, risks to human life and property are also highest in those regions.

A seismic hazard map for the U.S. clearly shows that the Pacific Coast of the western U.S. and Alaska has high risk paralleling the plate boundary.

However, there are also high-risk locations deep in plate interiors.

For the U.S. these include Hawaii and an area near New Madrid, Missouri which are associated with a volcanic hot spot and an ancient continental rift zone, respectively.

English

G4.3

Stress Release in Earthquakes: Elastic Rebound

After the 1906 earthquake along the San Andreas fault that destroyed San Francisco, one of the geologists who studied the event, Henry Fielding Reid, developed the elastic rebound theory to explain ground movement on active faults. He proposed that stress -- the local force per unit area -- builds up along a fault when the fault is locked. When the rocks fail and the fault ruptures, the stress is reduced as energy is released. Over time, stress along the fault can rebuild, eventually causing another earthquake, and continuing the cycle of earthquakes.

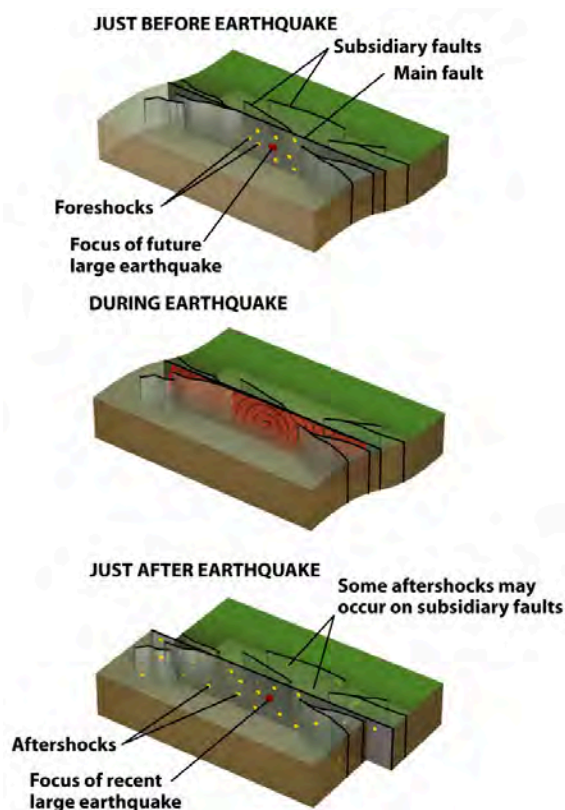


Figure 13.6
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Elastic rebound explains generally why earthquakes occur. In fact, earthquake events are much more complicated than the simple idea of stress buildup and release, usually consisting of multiple slip events that generate "foreshocks" and "aftershocks" to the main earthquake. It is important to keep in mind that faults are not lines on the surface, but planes of breakage underground, often consisting of multiple, overlapping segments that create "fault zones".

The amount of displacement on a fault -- the slip -- is directly related to the amount of energy released that creates seismic waves in an earthquake.

3

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G4.4

Earthquake Basics

Read

1. US Geological Survey. **"The Science of Earthquakes."** 

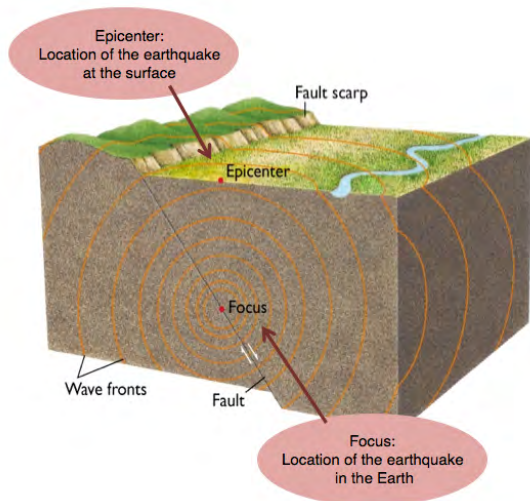
<http://earthquake.usgs.gov/learn/kids/eqscience.php>

This page covers the basic science of earthquakes. Be sure you understand how and why earthquakes occur.

Study Aids

1. If you come across terms in your reading and study that are unfamiliar the US Geological Survey's webpage, **Animations for Earthquake Terms and Concepts**  (https://www.usgs.gov/natural-hazards/earthquake-hazards/science/earthquake-animations?qt-science_center_objects=7#qt-science_center_objects) has definitions and small visualizations to explain them.

2. Interesting facts about earthquakes are compiled by the US Geological Survey. See **"Earthquake Facts"**  (https://www.usgs.gov/natural-hazards/earthquake-hazards/science/cool-earthquake-facts?qt-science_center_objects=0#qt-science_center_objects)."



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The Science of Earthquakes

By [Earthquake Hazards Program](#)

Originally written by Lisa Wald (U.S. Geological Survey) for “The Green Frog News”

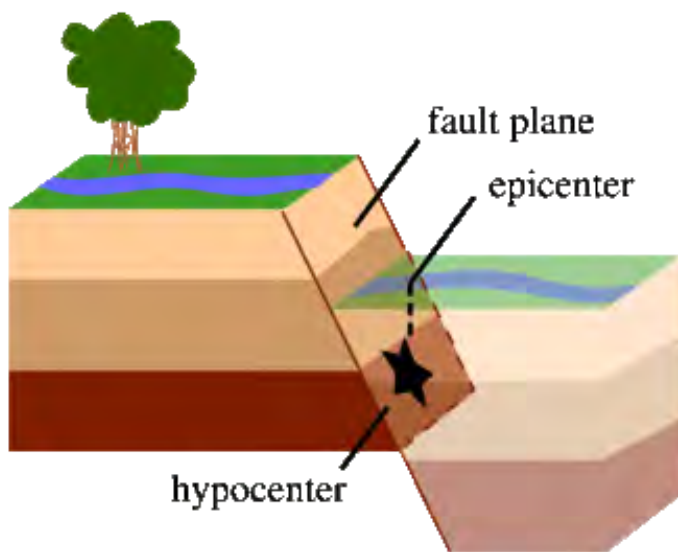
What is an earthquake?

An **earthquake** is what happens when two blocks of the earth suddenly slip past one another. The surface where they slip is called the **fault** or **fault plane**. The location below the earth’s surface where the earthquake starts is called the **hypocenter**, and the location directly above it on the surface of the earth is called the **epicenter**.

Sometimes an earthquake has **foreshocks**. These are smaller earthquakes that happen in the same place as the larger earthquake that follows. Scientists can’t tell that an earthquake is a foreshock until the larger earthquake happens. The largest, main earthquake is called the **mainshock**.

Mainshocks always

have **aftershocks** that follow. These are smaller earthquakes that occur afterwards in the same place as the mainshock. Depending on the size of the mainshock, aftershocks can continue for weeks, months, and even years after the mainshock!

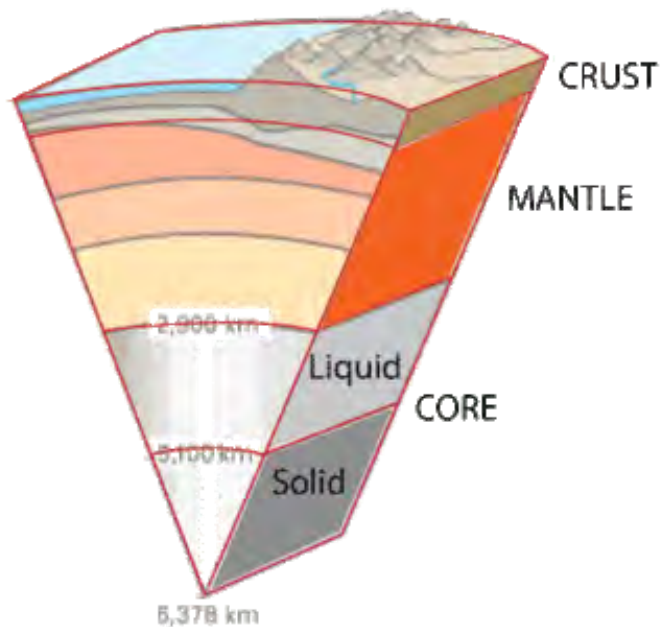


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A normal (dip-slip) fault is an inclined fracture where the rock mass above an inclined fault moves down (Public domain.)

Was this page helpful?

What causes earthquakes and where do they happen?



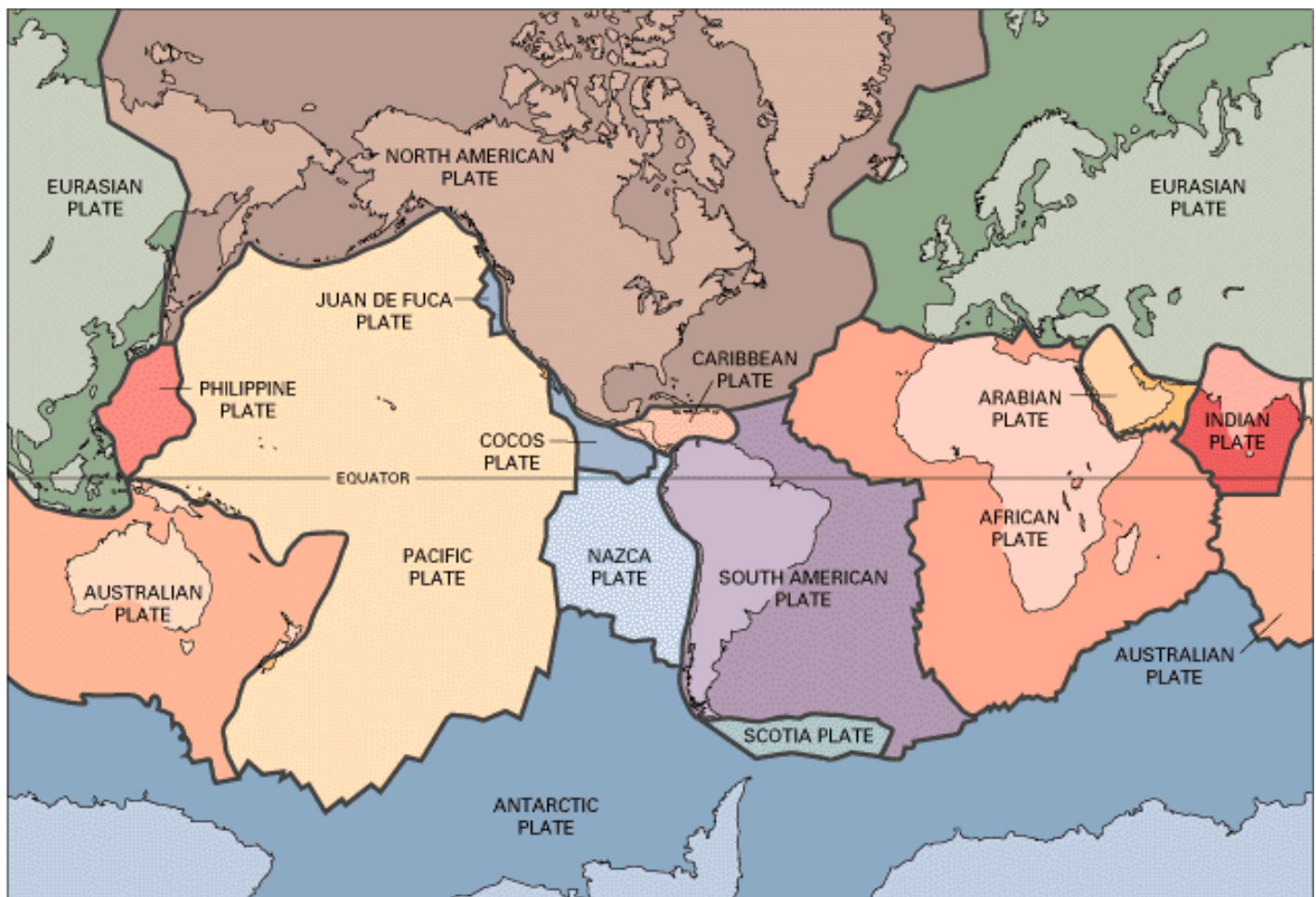
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A simplified cartoon of the crust (brown), mantle (orange), and core (liquid in light gray, solid in dark gray) of the earth. (Public domain.)

The earth has four major layers: the inner core, outer core, mantle and crust. The crust and the top of the mantle make up a thin skin on the surface of our planet.

But this skin is not all in one piece – it is made up of many pieces like a puzzle covering the surface of the earth. Not only that, but these puzzle pieces keep slowly moving around, sliding past one another and bumping into each other. We call these puzzle pieces **tectonic plates**, and the edges of the plates are called the **plate boundaries**. The plate boundaries are made up of many faults, and most of the earthquakes around the world occur on these faults. Since the edges of the plates are rough, they get stuck while the rest of the plate keeps moving. Finally, when the plate has moved far enough, the edges unstuck on one of the faults and there is an earthquake.

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The tectonic plates divide the Earth's crust into distinct "plates" that are always slowly moving. Earthquakes are concentrated along these plate boundaries. (Public domain.)

Why does the earth shake when there is an earthquake?

While the edges of faults are stuck together, and the rest of the block is moving, the energy that would normally cause the blocks to slide past one another is being stored up. When the force of the moving blocks finally overcomes the **friction** of the jagged edges of the fault and it unsticks, all that stored up energy is released. The energy radiates outward from the fault in all directions in the form of **seismic waves** like ripples on a pond. The seismic waves shake the earth as they move through it, and when the waves reach the earth's surface, they shake the ground and anything on it, like our houses and us!

How are earthquakes recorded?

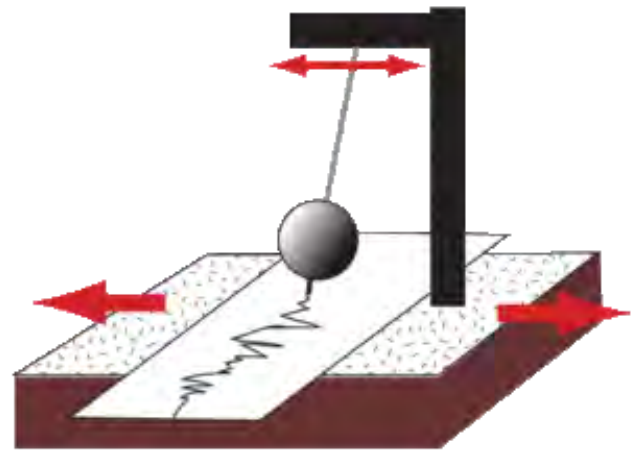
Earthquakes are recorded by instruments called **seismographs**. The recording they make is called a **seismogram**. The seismograph has a base that sets firmly in the ground, and a heavy weight that hangs free. When an earthquake causes the ground to shake, the base of the seismograph shakes too, but the hanging weight

Was this page helpful? [Log or](#)

string that it is hanging from absorbs all the movement. The difference in position between the shaking part of the seismograph and the motionless part is what is recorded.

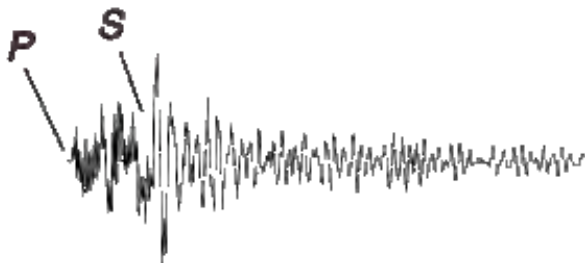
How do scientists measure the size of earthquakes?

The size of an earthquake depends on the size of the fault and the amount of slip on the fault, but that's not something scientists can simply measure with a measuring tape since faults are many kilometers deep beneath the earth's surface. So how do they measure an earthquake? They use the **seismogram** recordings made on



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The cartoon sketch of the seismograph shows how the instrument shakes with the earth below it, but the recording device remains stationary (instead of the other way around). (Public domain.)



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An example of a seismic wave with the P wave and S wave labeled. (Public domain.)

for each earthquake. Scientists also talk about the *intensity* of shaking from an earthquake, and this varies depending on where you are during the earthquake.

How can scientists tell where the earthquake happened?

Seismograms come in handy for locating earthquakes too, and being able to see the **P wave** and the **S wave** is important. You learned how P & S waves each shake the ground in different ways as they travel through it. P waves are also faster than S waves, and this fact is what allows us to tell where an earthquake was. To understand how this works, let's compare P and S waves to lightning and thunder.

Was this page helpful? ☐ Yes ☐ No

during a thunderstorm you will first see the lightning and then you will hear the thunder. If you are close to the lightning, the thunder will boom right after the lightning, but if you are far away from the lightning, you can count several seconds before you hear the thunder. The further you are from the storm, the longer it will take between the lightning and the thunder.

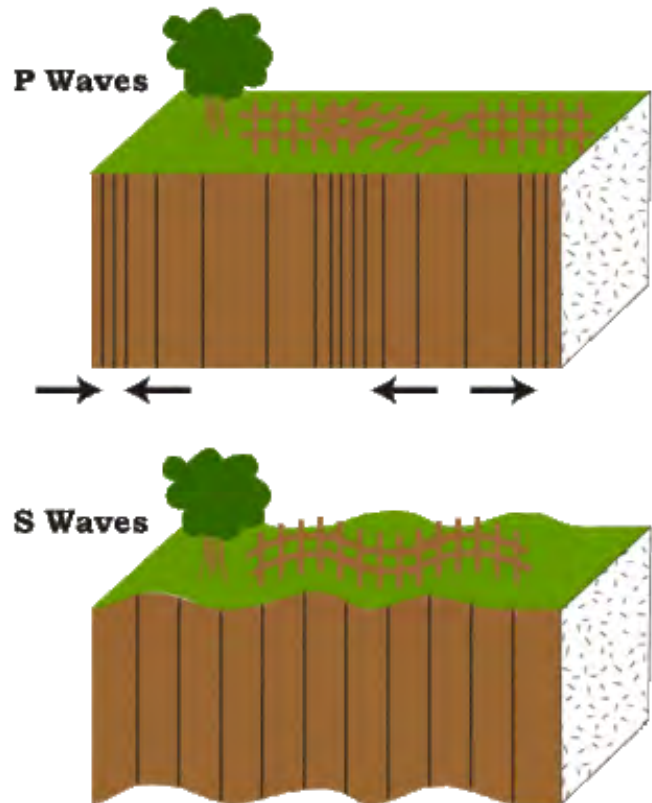
P waves are like the lightning, and S waves are like the thunder. The P waves travel faster and shake the ground where you are first. Then the S waves follow and shake the ground also. If you are close to the earthquake, the P and S wave will come one right after the other, but if you are far away, there will be more time between the two.

By looking at the amount of time between the P and S wave on a seismogram recorded on a seismograph, scientists can tell how far away the earthquake was from that location. However, they can't tell in what direction from the seismograph the earthquake was, only how far away it was. If they draw a circle on a map around the station where the **radius** of the circle is the determined distance to the earthquake, they know the earthquake lies somewhere on the circle. But where?

Scientists then use a method called **triangulation** to determine exactly where the earthquake was (see image below). It is called triangulation because a triangle has three sides, and it takes three seismographs to locate an earthquake. If you draw a circle on a map around three different seismographs where the **radius** of each is the distance from that station to the earthquake, the intersection of those three circles is the **epicenter**!

Can scientists predict earthquakes?

No, and it is unlikely they will ever be able to predict them. Scientists have tried many different ways of predicting earthquakes, but none have



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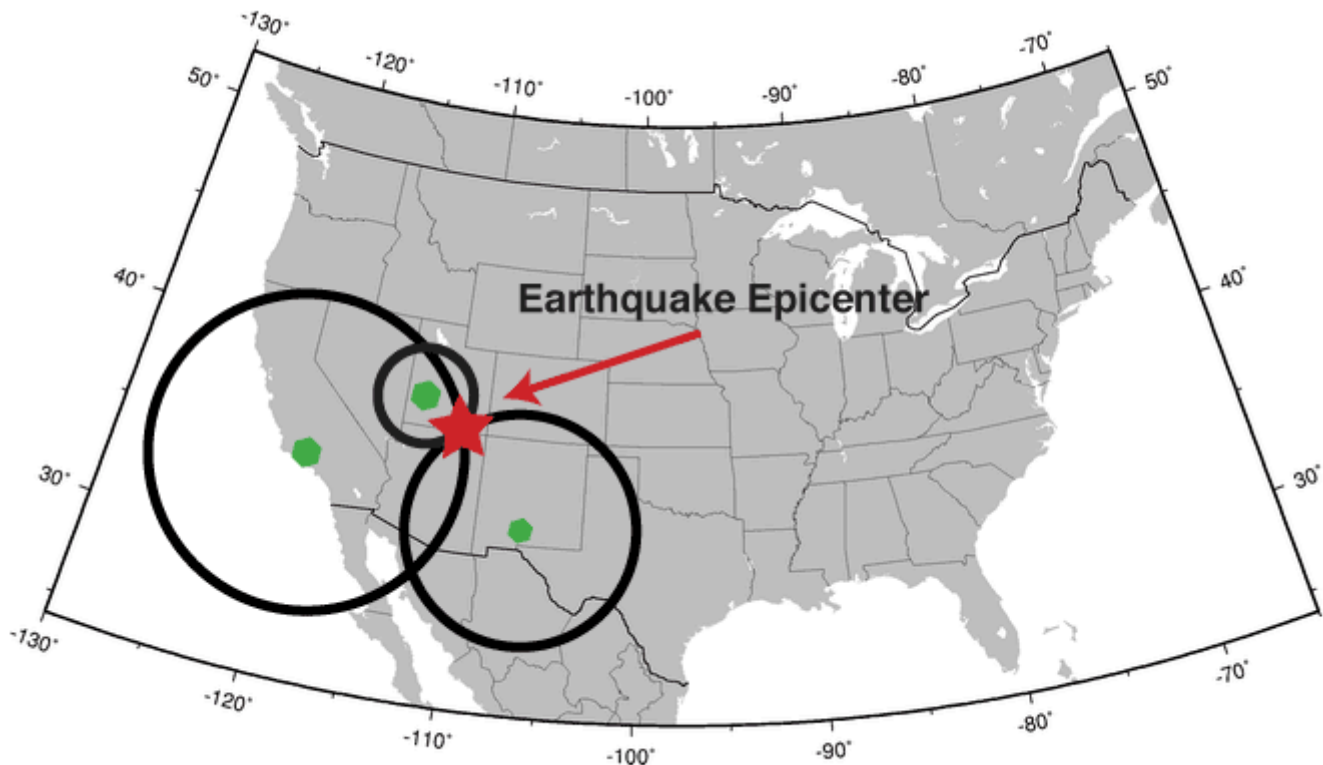
P Waves alternately compress and stretch the crustal material parallel to the direction they are propagating. S Waves cause the crustal material to move back and forth perpendicular to the direction they are travelling. (Public domain.)

Was this page helpful?

particular fault, scientists know there will be another earthquake sometime in the future, but they have no way of telling when it will happen.

Is there such a thing as earthquake weather? Can some animals or people tell when an earthquake is about to hit?

These are two questions that do not yet have definite answers. If weather does affect earthquake occurrence, or if some animals or people can tell when an earthquake is coming, we do not yet understand how it works.



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Triangulation can be used to locate an earthquake. The seismometers are shown as green dots. The calculated distance from each seismometer to the earthquake is shown as a circle. The location where all the circles intersect is the location of the earthquake epicenter. (Public domain.)

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U.S. Geological Survey
U.S. Department of the Interior



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G4.5

Seismic Waves

When a fault breaks suddenly, stress is released in the form of intense vibrations ("shock" waves) that travel outward from a source (the focus) of energy. Seismic waves are caused by the release of energy that may occur, for example, from earthquakes, explosions, or other anthropogenic sources.

Types of Seismic Waves:

- ◆ P-waves: Primary waves – compressional waves
- ◆ S-waves: Secondary waves – shear wave
- ◆ Surface waves: Love or Rayleigh waves – two types of waves that travel on the Earth's surface

View

Animations of seismic wave motions, and more in depth information about how seismic waves propagate:

1. [Types of seismic waves are described \[2:30\]](https://www.youtube.com/watch?v=xzYnJr1fIB4)  (<https://www.youtube.com/watch?v=xzYnJr1fIB4>)
2. [Simple animation wave motions](https://www.edumedia-sciences.com/en/media/426-the-3-types-of-seismic-waves)  (<https://www.edumedia-sciences.com/en/media/426-the-3-types-of-seismic-waves>)  (https://www.iris.edu/hq/inclass/animation/seismic_wave_motions?zoombox=7)
3. [Simulation of Wave motions through the Earth's interior](https://www.iris.edu/hq/inclass/software-web-app/seismic_waves_viewer)  (https://www.iris.edu/hq/inclass/software-web-app/seismic_waves_viewer)

The following pages provide additional details, links, and visualizations of seismic wave motions and properties.

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A quick glance at the distribution of large earthquake epicenters immediately shows that their locations are not randomly distributed over the globe.

They are highly concentrated along linear to arcuate bands, particularly adjacent to the edges of continents and along mid-ocean ridges. These bands mark the edges of the tectonic plates.

Quakes are common at all three types of plate boundaries.

At divergent boundaries, tensional forces break up the lithosphere along normal faults.

This results in abundant shallow-focus earthquakes.

At convergent boundaries, dense oceanic plates descend into the mantle.

This leads to shallow, intermediate, and deep-focus quakes as the plate makes its way downward at an angle.

The deepest focus quakes will therefore have epicenters well into the overriding plate, such as in western South America.

Transform boundaries have numerous shallow to intermediate focus quakes as lithospheric plates slide past one another.

The San Andreas fault is the best known example of this.

Because earthquake activity is so concentrated along plate boundaries, risks to human life and property are also highest in those regions.

A seismic hazard map for the U.S. clearly shows that the Pacific Coast of the western U.S. and Alaska has high risk paralleling the plate boundary.

However, there are also high-risk locations deep in plate interiors.

For the U.S. these include Hawaii and an area near New Madrid, Missouri which are associated with a volcanic hot spot and an ancient continental rift zone, respectively.

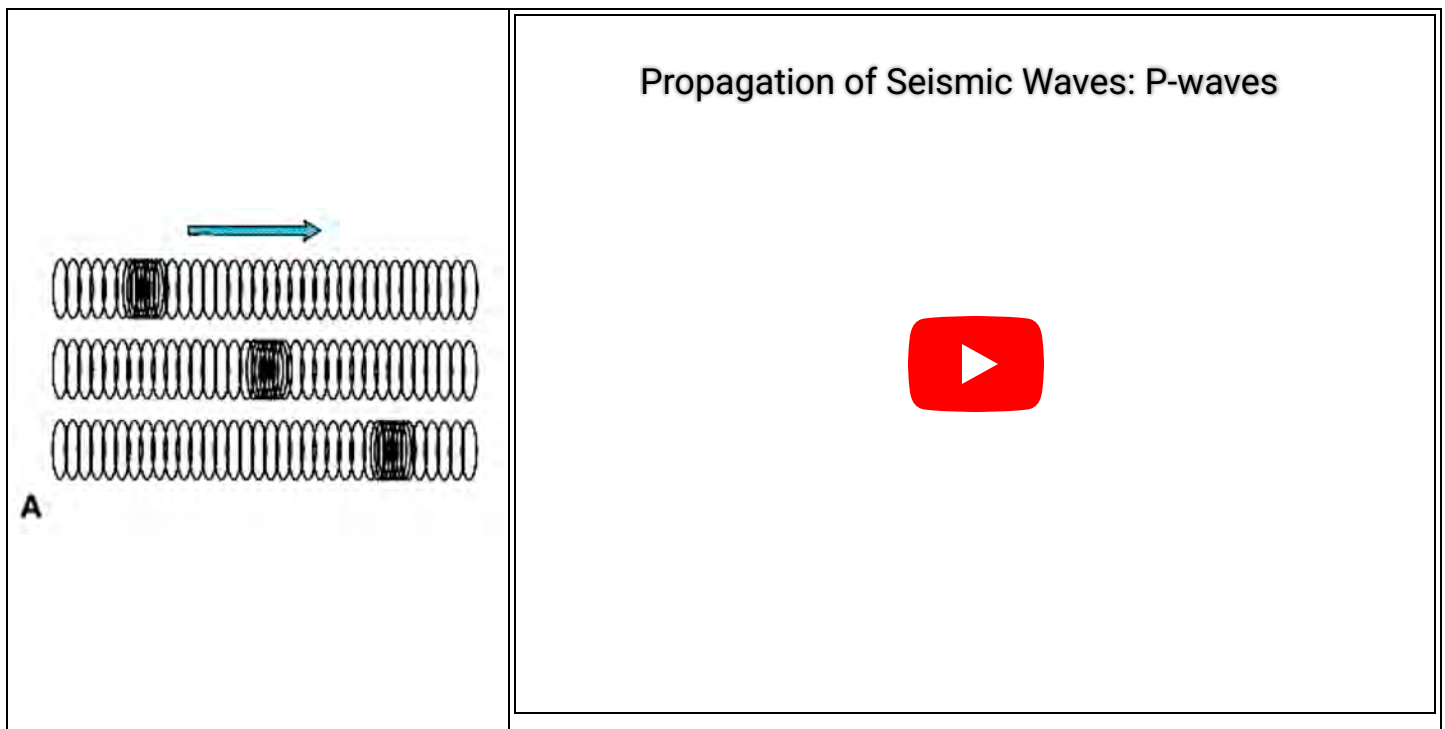
English

G4.6

Properties of P-Waves (primary waves)

Compressional seismic waves (similar to sound waves):

- ◆ Move by compressing and expanding material (push-pull) in the direction the wave is traveling
- ◆ Travel at the fastest speed through the Earth of all types of seismic waves
- ◆ Travel faster through solid rock than sound waves, moving at about 6-8 km/second
- ◆ Can travel through both solids and liquids



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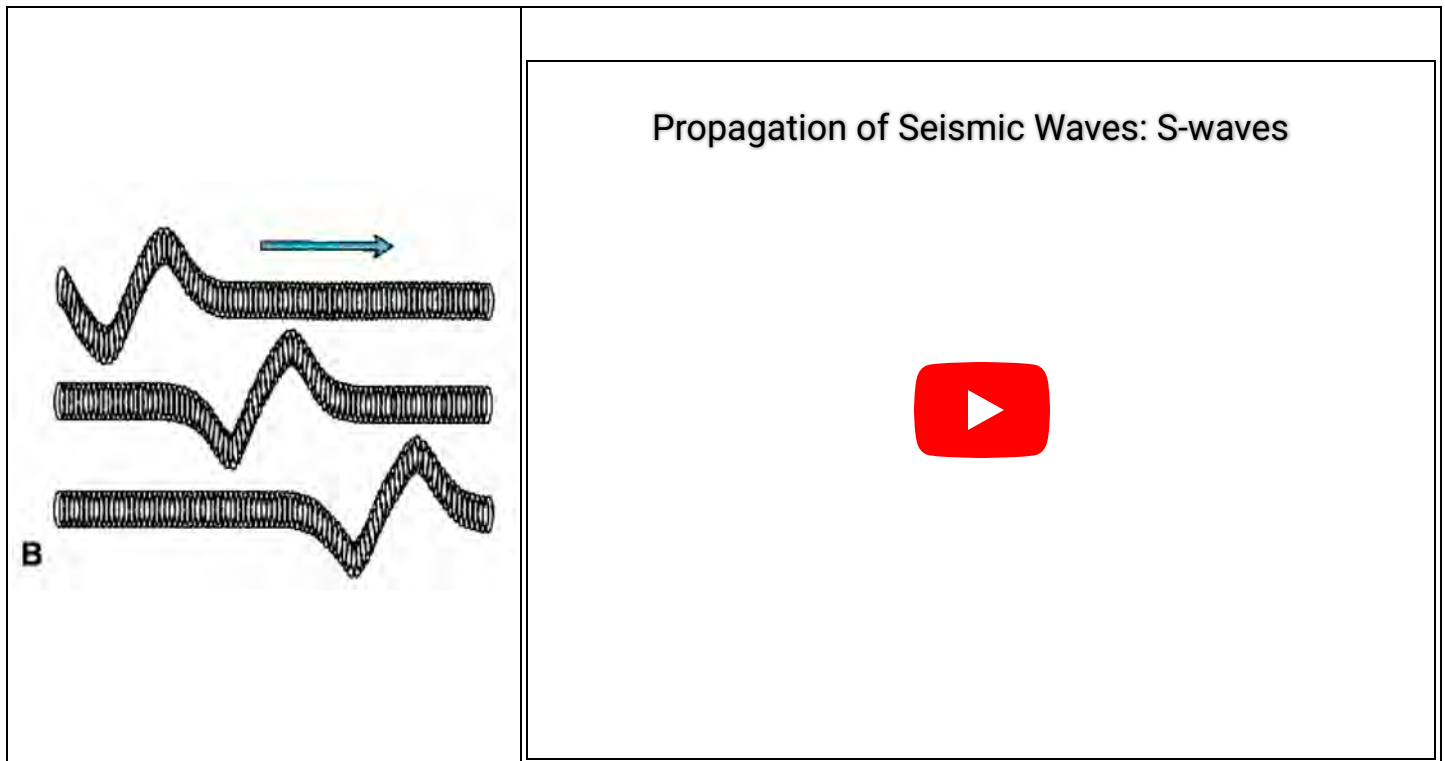
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G4.7

Properties of S-Waves (secondary waves)

Shear seismic waves (back/forth, up/down motion):

- ◆ Wave motion is perpendicular to the direction of wave travel
- ◆ Travel at slower speeds through the Earth than P-waves, so they arrive after P-waves
- ◆ Travel at about half of the speed of P-waves through rock, at about 3-4 km/second
- ◆ Can travel **only** through solids, **not** through liquids



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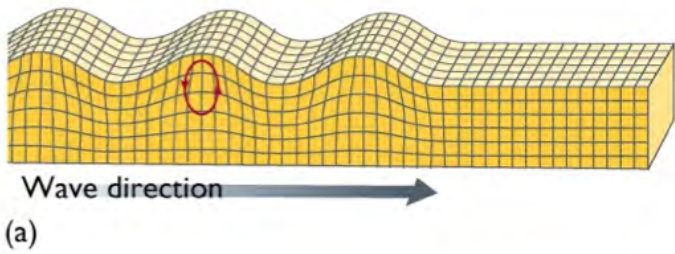
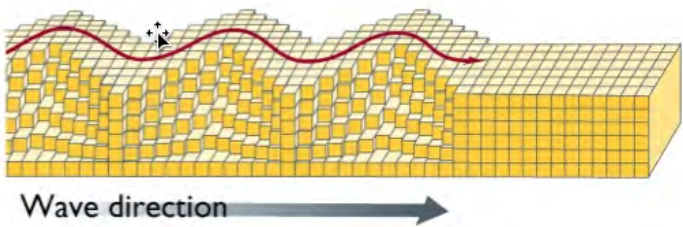
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G4.8

Properties of Surface Waves

Surface waves travel **only** on the surface of the Earth:

- ◆ Their speed is less than that of S-waves
- ◆ Motion is either elliptical, similar to that of ocean waves (Rayleigh waves), or
- ◆ Motion is perpendicular (back-forth) to direction of wave travel (Love waves)

 (a)	Propagation of Seismic Wa... 
 (b)	Propagation of Seismic Wa... 

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G4.9

Measuring Earthquakes: The Seismograph and Magnitude Scales

Seismograph designed to detect vertical movement

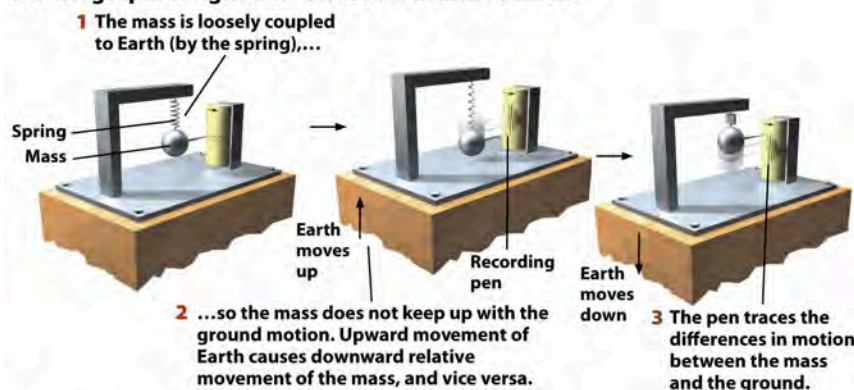


Figure 13.7a
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The magnitude of an earthquake is a measure of its size. After the invention of accurate seismographs in the early 20th century to record the motion from seismic waves, Charles Richter created a numerical scale in 1935 to gauge earthquake size using the motion measured on a seismogram.

He used a logarithmic scale based on the maximum displacement or amplitude of the earthquake signal as seen on the seismogram, with a correction for the distance between the focus and the seismometer. This was necessary because the closer the seismometer is to the earthquake, the larger the amplitude on the seismogram, irrespective of the size or magnitude of the event (read on...).



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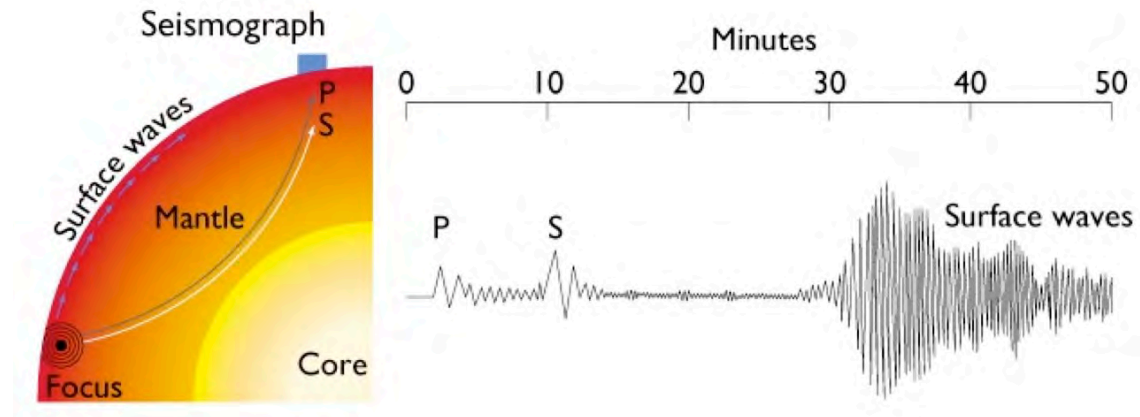
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G4.10

Measuring Earthquakes: The Seismograph and Magnitude Scales (continued)

Although we still hear about the "Richter scale" for earthquakes, seismologists today use magnitude scales that relate directly to the amount of motion along the fault and other physical characteristics of the rupture zone to determine the size of an earthquake. Numerically, however, earthquake magnitudes calculated by these newer methods are roughly equivalent to Richter's original scale. The Richter local magnitude (ML) is defined to be used for 'local' earthquakes up to 600 km away.



It is important to note that earthquake magnitude scales are logarithmic, which means that **each increment** on the scale corresponds to an earthquake that is **10 times larger**. For example, an earthquake of magnitude 5 is ten times larger than a magnitude 4 earthquake, and 100 times larger than a magnitude 3 earthquake.

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G4.11

Determining Earthquake Location and Magnitude

EQ_location_animation



This video animation [2:00] describes how to locate the epicenter of an earthquake from a single seismograph. In practice, seismologists use the arrival times of P- and S-waves from seismic stations all over the world to determine the location and magnitude of an earthquake.

For more information, see the [US Geological Survey FAQ: Measuring Earthquakes](https://www.usgs.gov/faqs/how-are-earthquakes-recorded-how-are-earthquakes-measured-how-magnitude-earthquake-determined?qt-news_science_products=0#qt-news_science_products). [↗](https://www.usgs.gov/faqs/how-are-earthquakes-recorded-how-are-earthquakes-measured-how-magnitude-earthquake-determined?qt-news_science_products=0#qt-news_science_products)
(https://www.usgs.gov/faqs/how-are-earthquakes-recorded-how-are-earthquakes-measured-how-magnitude-earthquake-determined?qt-news_science_products=0#qt-news_science_products)

Practice Problem

How many times larger is a magnitude 9 earthquake than a magnitude 6 earthquake?

Answer this on your own before checking the solution on the next page.

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Earthquakes release their energy in the form of vibrations known as seismic waves. These include P, S, and Surface waves.

Because each of these waves travels through rock at a different known velocity, P waves fastest, S waves more slowly, and Surface waves most slowly, the time interval between when one wave and the next arrive at a distant location can be used to determine the distance to the site of the quake.

The longer the interval between the arrival of the P and S waves at a given station, the farther the distance to the earthquake focus.

A time delay of 24 seconds would, for example, indicate a distance to the earthquake focus of just over 200 kilometers.

This distance, combined with the maximum displacement on a seismogram, can also be used to determine the Richter magnitude of the quake.

Because the time delay between P and S wave arrival gives the distance to an earthquake, a single seismogram can be used to draw on a map a circle around the seismograph station that has a radius equal to the calculated distance.

This gives only a distance, but not a direction or a unique location.

The earthquake epicenter could lie anywhere on the drawn circle.

However, if a seismogram from a second seismograph location is used, the two circles now drawn on the map will intersect in exactly two places, one of which is the epicenter location.

Using a third station, a unique point where all three circles overlap will be produced.

This is how earthquake epicenters are located.

G4.12

Practice Problem Solution

Question: *How many times larger is a magnitude 9 earthquake than a magnitude 6 earthquake?*

Solution

1. Convert $\log_{10}$ numbers to exponential form, as follows:

$$\text{Magnitude 9} = 10^9$$

$$\text{Magnitude 6} = 10^6$$

2. Subtract: $10^9 - 10^6 = 10^3$ *Hint: subtract the exponents after converting the log value*

Solution: = 1000 times larger

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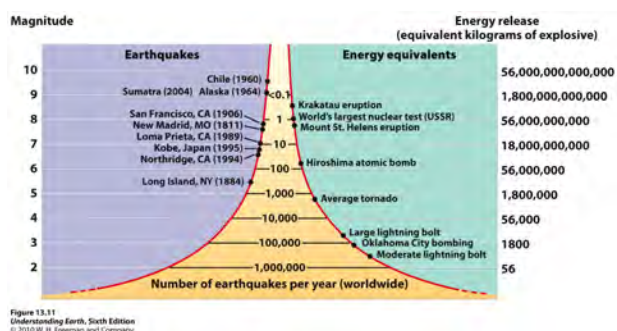
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G4.13

Earthquake Magnitude and Energy Release

For comparison purposes, a magnitude 5 ML earthquake is equivalent to the explosion of 1,000 tons of TNT whereas a magnitude 6 ML earthquake is the energy equivalent of 30,000 tons of TNT or a 30 kilotonne nuclear explosion. In terms of frequency, note that many more small earthquakes occur each year compared to large earthquakes.



For practice calculating differences in earthquake magnitudes, see the [US Geological Survey: Earthquake Magnitude Calculator](https://earthquake.usgs.gov/education/calculator.php). [↗](https://earthquake.usgs.gov/education/calculator.php)
(<https://earthquake.usgs.gov/education/calculator.php>)

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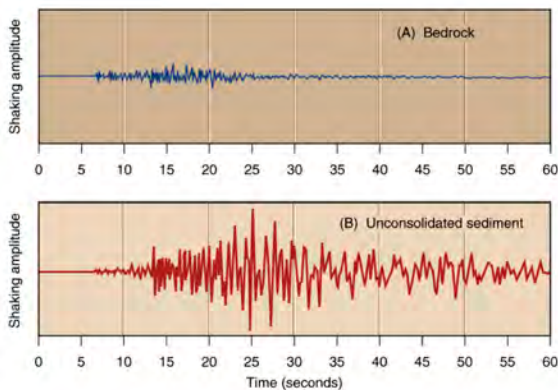
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G4.14

Shaking Intensity and Damage: What affects the shaking intensity you actually feel in an earthquake?

Most earthquake magnitude scales relate to the amount of energy released as rocks slip on fault planes and stress is reduced. What you feel depends of course on how far away you are, but also on other factors that influence how much the ground shakes at your particular location. These include:



- Distance from the earthquake focus
- Earthquake depth and seismic wave propagation through the Earth
- Rock type: soft, water-filled sediments shake more; hard bedrock shakes less
- Whether ground failure occurs (liquefaction, ground rupture)

In terms of impact on people, building and road construction practices are a critical factor in determining the amount of damage and loss of life from an earthquake. In most countries, the extent of human loss and injuries depends mostly on whether buildings can withstand a strong earthquake. In some large earthquakes, “triggered” effects such as tsunamis, landslides, or fires can be more devastating than the shaking from the earthquake. For example, the 2004 Indonesian earthquake triggered a devastating tsunami that swept across the Indian ocean, killing more than 200,000 people.



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G4.15

Modified Mercalli Intensity (MMI) Scale

TABLE 13.1 Modified Mercalli Intensity Scale

Intensity Level	Description
I	Not felt.
II	Felt only by a few people at rest. Suspended objects may swing.
III	Felt noticeably indoors. Many people do not recognize it as an earthquake. Parked cars may rock slightly.
IV	Felt indoors by many, outdoors by few. Dishes, windows, doors rattle. Parked cars rock noticeably.
V	Felt by most; many awakened. Some dishes, windows broken. Unstable objects overturned.
VI	Felt by all. Some heavy furniture moves. Damage slight.
VII	Slight to moderate damage in well-built structures; considerable damage in poorly built structures; some chimneys broken.
VIII	Considerable damage in well-built structures. Damage great in poorly built structures. Fall of chimneys, factory stacks, columns, monuments, walls.
IX	Damage great in well-built structures, with partial collapse. Buildings shifted off foundations.
X	Some well-built wooden structures destroyed; most masonry and frame structures destroyed. Rails bent.
XI	Few if any masonry structures remain standing. Bridges destroyed. Rails bent greatly.
XII	Damage total. Lines of sight and level are distorted. Objects thrown into the air.

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A scale of felt shaking intensity, called the modified Mercalli Intensity (MMI) Scale has been developed based on observations of what people feel and observe during an earthquake. This scale is useful for evaluating the magnitude of earthquakes described in historical records before seismographs and for determining local impacts from an earthquake that has just occurred.

"Did You Feel It?"

(<http://earthquake.usgs.gov/earthquakes/dyfi/>)

is the US Geological Survey's website for people to enter information about what they felt during an earthquake. Based on the responses gathered from this site, the USGS creates a "shake map" of the affected area using the Modified Mercalli Intensity Scale.

This concludes Part 1 of Geo3.

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G4P2.1

Volcanoes

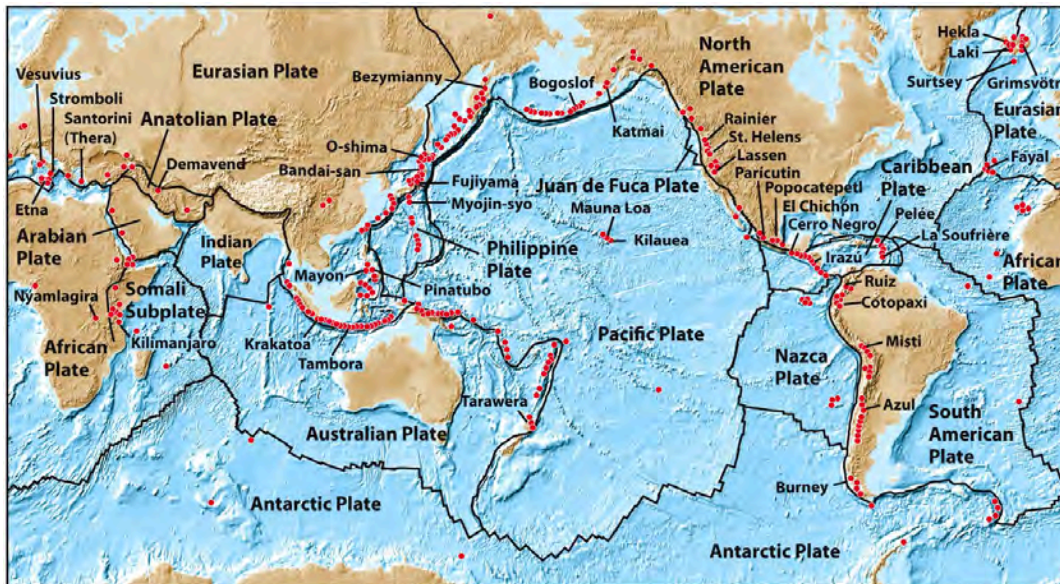


Figure 12.24
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Similar to earthquakes, the location of active volcanoes and their styles of eruption are explained well within the framework of plate tectonic theory. Volcanoes differ from earthquakes, however, in that the generation and movement of magma in the Earth prior to a major eruption can often be detected, and eruptions may last from days to months rather than occurring for only seconds to minutes.

View

Watch this video animation, [Volcanism and Plate Tectonics](http://youtu.be/pPDqbwIXl8I) (<http://youtu.be/pPDqbwIXl8I>), that describes the relationship between volcanoes and plate tectonics. Note the distribution of active volcanoes, their relationship to melting of crust and generation of magmas, and the special case of a “hot spot” volcano – the Hawaiian Islands.

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A quick glance at the distribution of active volcanoes above sea level immediately shows that their locations are not randomly distributed over the globe. They are highly concentrated along linear to arcuate bends, particularly adjacent to the edges of continents.

Most of the rest of the active volcanoes are found on the sea floor in similar linear to arcuate bends.

These bends mark the edges of the tectonic plates.

Volcanism is common at both divergent and convergent plate boundaries, with magma being generated in large quantities at both locations.

At divergent boundaries, magma is generated primarily by decompression melting of already hot rocks at shallow depths, along mid-ocean ridges.

At convergent boundaries, melting is produced primarily by temperature increases in water-rich rocks, as they descend into the mantle at the subduction zone.

While the location of most of earth's active volcanoes outlines major plate boundaries, hot spot volcanoes in the interior of the plates allow us to track the direction and speed of plate motions over time.

Hot spots appear to be the result of rising plumes of hot material from deep within the mantle, probably originating at the core mantle boundary.

As magma bodies from hot spots periodically puncture the surface, volcanoes are produced,

which are then carried away from the hot spot by plate motions.

By dividing the distance of an old volcano from the active hot spot by the age of the volcano, the average speed of plate motion over that time can be determined.

The direction of plate motion can then be deduced by the orientation of the resulting chain of volcanoes tracing back to the active hotspot.

Examples include the Hawaii Emperor chain in the Pacific and the string of calderas running from Southeast Oregon to Yellowstone in Northwest Wyoming.

Understanding the location and type of plate boundaries and the volcanoes that mark them, make it possible to better assess the potential hazards to human life and property that future eruptions may represent.

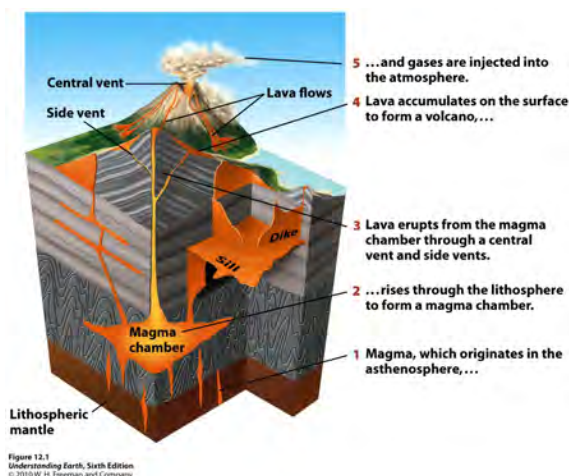
By mapping out the types and ages of eruptions, the timeframe and size of the erupted hazards can be evaluated.

The ages and positions of volcanoes in the western US show that, aside from the hotspot in Hawaii, volcanic activity has been confined almost entirely to the west coast and Alaska, over the past million years. Indeed, all but one eruption in the past 300 years occurred either in Hawaii or along the convergent boundary running from Northern California to the end of the Aleutian Island chain.

G4P2.2

Volcanoes: Agents of Transport from the Earth's Interior

Volcanoes are eruptions of molten rock (magma) that rise from the upper mantle or crust and break through the surface. There are two ways to make magma:



1. Subducting lithosphere to sufficient depth such that heat and pressure melt rocks
2. Reducing pressure on rocks in the upper mantle, which melts them and enables the magma to rise to the surface at, for example, divergent plate boundaries

The style of a volcanic eruption, whether the erupted material flows as molten rivers of lava or explodes violently into the air, is determined by the composition of the magma and the concentration of volatile gases that are dissolved in it.

2

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G4P2.3

Styles of Volcanic Eruptions

Why do some volcanoes explode violently, with the ejection of large plumes of ash and pyroclastic material, while other volcanic eruptions produce free-flowing lavas?

Two important factors that determine whether volcanoes erupt violently are:

1. The amount of gas trapped in the molten magma under pressure. In the Earth, gases are dissolved in magma and held there under pressure. When magmas rise in the crust and approach the surface, pressure is reduced and gases are released – either slowly or explosively. The primary gas dissolved in magma is water vapor (H_2O), which typically accounts for most of the volatile gas (about 70-95%). Other common gases in magmas are carbon dioxide (CO_2), sulfur dioxide (SO_2), and small amounts of nitrogen, hydrogen, helium, and carbon monoxide.





Figure 12.2
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2. The viscosity of the magma – how freely it flows – is determined by its composition. Magmas of mafic composition (i.e., basalt) are less viscous compared to magmas of felsic composition (i.e., andesite or rhyolite). Low viscosity magmas allow trapped gases to escape as pressure is released during an eruption. Very thick, viscous lavas trap gases and cause pressure to build until the gases explode as the volcano erupts.

Overall, gases make up only a small percent of the mass of the magma, but they have a large effect on the amount of ash and volcanic material that explodes at the surface.

3

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G4P2.4

Volcanic Hazards

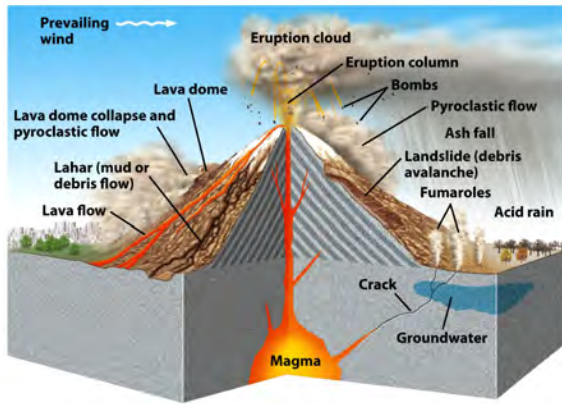


Figure 12.29
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Unlike earthquakes, active volcanoes typically issue warning signs before a major eruption. These signs include earthquake tremors that signal the movement of magma at depth, bulges or elevation change at the surface (inflation), and changes in gas chemistry. In addition, active volcanoes typically erupt over periods of decades to thousands of years. Therefore, documenting the history and extent of historic volcanic eruptions can provide clues about the size of the next eruption, and hazard zones can be determined to prepare people who may be in danger in the event of an eruption.

View

The US Geological Survey has produced a series of [Volcano Web Shorts](https://www.usgs.gov/products/multimedia-gallery/videos)  (<https://www.usgs.gov/products/multimedia-gallery/videos>) that describe how volcanoes are studied and monitored, and how this work serves to prepare for impacts from volcanic eruptions. Go to the web site and view each of the short animations listed below (2-3 minutes each):

1. [Photogrammetry](https://www.usgs.gov/media/videos/volcano-web-shorts-1-photogrammetry)  (<https://www.usgs.gov/media/videos/volcano-web-shorts-1-photogrammetry>)
2. [Debris Flows](https://www.usgs.gov/media/videos/volcano-web-shorts-2-Debris-Flows)  (<https://www.usgs.gov/media/videos/volcano-web-shorts-2-Debris-Flows>)
3. [Seismology](https://www.usgs.gov/media/videos/volcano-web-shorts-3-Seismology)  (<https://www.usgs.gov/media/videos/volcano-web-shorts-3-Seismology>)
4. [Instruments](https://www.usgs.gov/media/videos/volcano-web-shorts-4-Instruments)  (<https://www.usgs.gov/media/videos/volcano-web-shorts-4-Instruments>)
5. [Volcanic Ash Impacts](https://www.usgs.gov/media/videos/volcano-web-shorts-5-Volcanic-Ash-Impacts)  (<https://www.usgs.gov/media/videos/volcano-web-shorts-5-Volcanic-Ash-Impacts>)
6. [Societal Impacts of Volcanism](https://www.usgs.gov/media/videos/volcano-web-shorts-6-societal-impacts-volcanism)  (<https://www.usgs.gov/media/videos/volcano-web-shorts-6-societal-impacts-volcanism>)

For each video determine:

1. What is being studied and why is it important?
2. How does this information help people understand and prepare for volcanic eruptions?

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G4P2.5



Summary

Be sure you study and review the following:

Earthquakes

- Be able to relate earthquake generation, location, depth, and size to processes at tectonic plate boundaries
- Understand the types of seismic waves generated by earthquakes
- Know that earthquake magnitude scales are logarithmic scales, and know how to interpret differences in earthquake magnitudes

Volcanoes

- Be able to describe how plate tectonic processes relate to the origin of magmas and volcanic eruptions
- Understand how gases in magmas contribute to the explosiveness of a volcanic eruption

For both

- Be able to describe the impacts on humans from earthquakes and volcanoes, and how we can prepare for these hazards

This concludes Part 2 of the Geo3 eBook

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(<https://catcourses.ucmerced.edu/courses/32116/pages/geo4-plate-tectonics-earthquakes-and-volcanoes>).

G4P2.6

Volcanoes and Earthquakes Study Case:

Fire and Ice: locating earthquakes in an active volcano on Iceland

Background

Volcanic eruptions can occur in a variety of ways, some relatively benign and some catastrophically dangerous. In 2010 the subglacial Eyjafjallajökull eruption on Iceland caused a large outpouring of steam and ash into the upper atmosphere, severely disrupting European airspace for nearly two weeks. This is one of the reasons why scientists are continually monitoring volcanos on Iceland to try and determine when and how they are likely to erupt next.

In 2014 scientists from the University of Cambridge volcano seismology group were monitoring seismic activity in the area surrounding the Vatnajökull ice cap in Iceland using a network of 72 seismometers.



The temporary seismic stations were recovered before being

Some seismic stations were even installed on top of the ice cap

A fissure eruption is a spectacular sight

g over-run by lava flows

The eruption

In August 2014 seismic activity in the subglacial Bardarbunga volcano increased dramatically and by 17th August it was apparent that a magma intrusion was heading northeast from the volcano at a depth of 5-7 km below the surface.

By the end of August this underground molten magma had moved 46 km and eventually reached the surface with a spectacular fissure eruption. Fortunately this eruption occurred at Holuhraun, an area just outside the icecap and before reaching the large magma chamber beneath the Askja volcano.

An eruption beneath the icecap could have caused an explosive eruption similar to the 2010 Eyjafjallajökull event or if the intrusion had linked up with the Askja magma chamber it might have triggered an enormous eruption similar to that of 1875 which devastated northern Iceland.

The eruption in Holuhraun continued for over 6 months erupting a total of 1.4 km³ of lava over an area of 84 km², it was an impressive spectacle but did not cause large scale devastation.

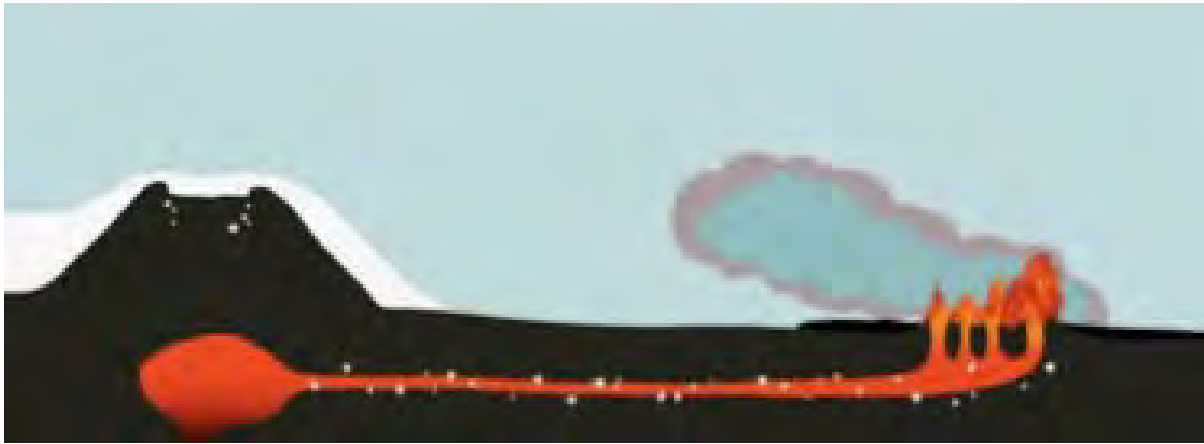


Figure 1: Molten magma can travel large distances before erupting to the surface

The scientific results

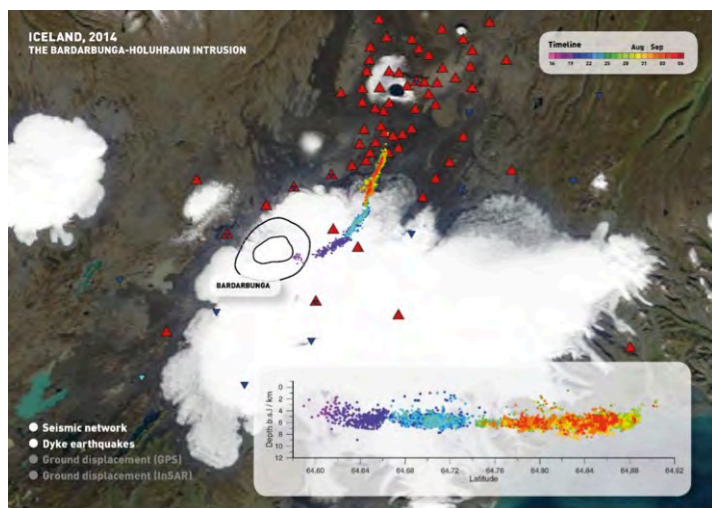
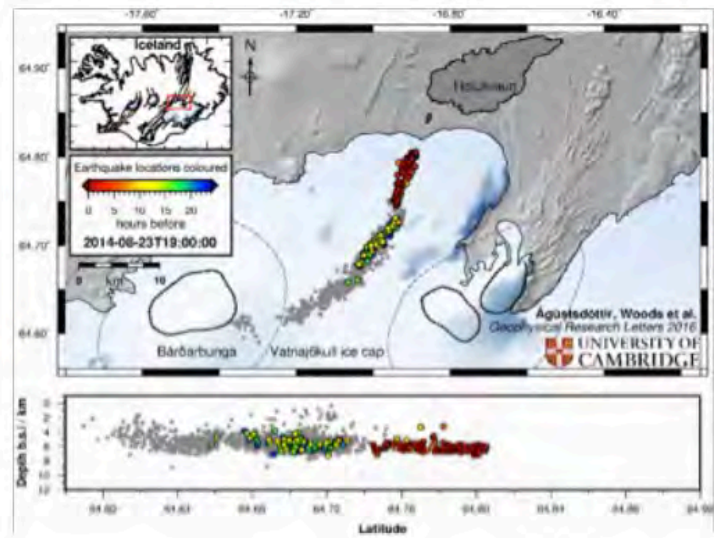
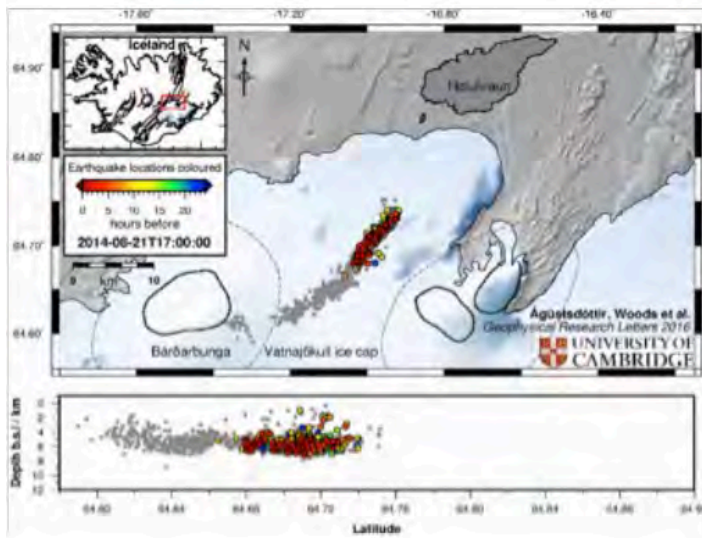
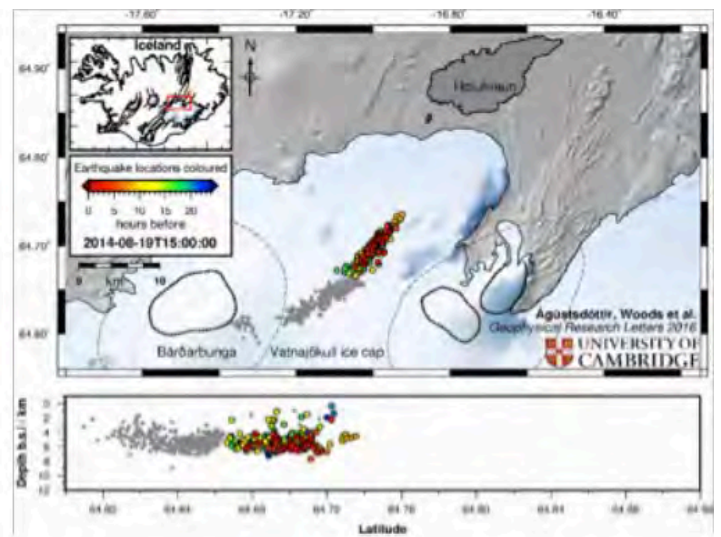
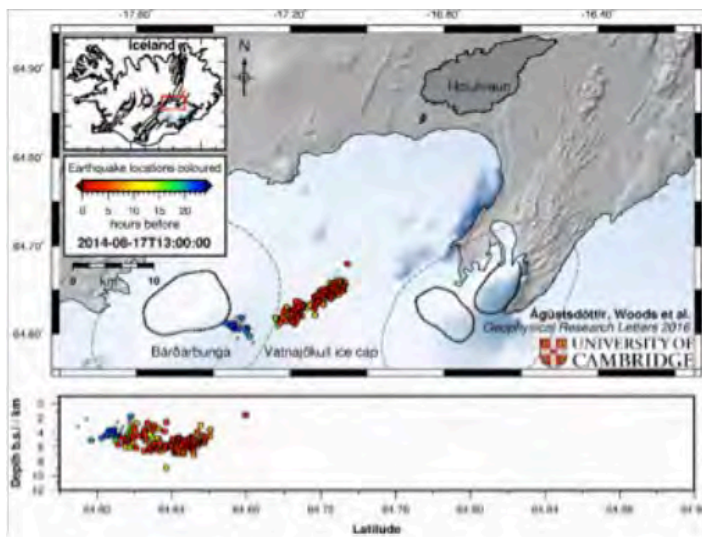


Figure 2: Location map of Cambridge seismometers in August 2014 with earthquake locations and dates

The data collected by these seismometers has allowed scientists in Cambridge to map the underground movement of magma from Bardarbunga to Holuhraun in unprecedented detail and is leading to a greater understanding of how the plumbing systems beneath volcanos operate. In total over 30,000 earthquakes with magnitudes ranging from approximately 0.5 to 4 were located during this particular volcanic episode. Most of the seismic activity was concentrated at the tip of the magma intrusion, 5-7 km beneath the surface, advancing at 0.3-4.7 km/h with stationary intervals.

Being able to map the time progression of underground magma movements on a minute by minute and hour by hour basis with such precision was only made possible by the fortunate coincidence of Cambridge scientists having such a large number of seismometers already in the right place when the activity commenced.



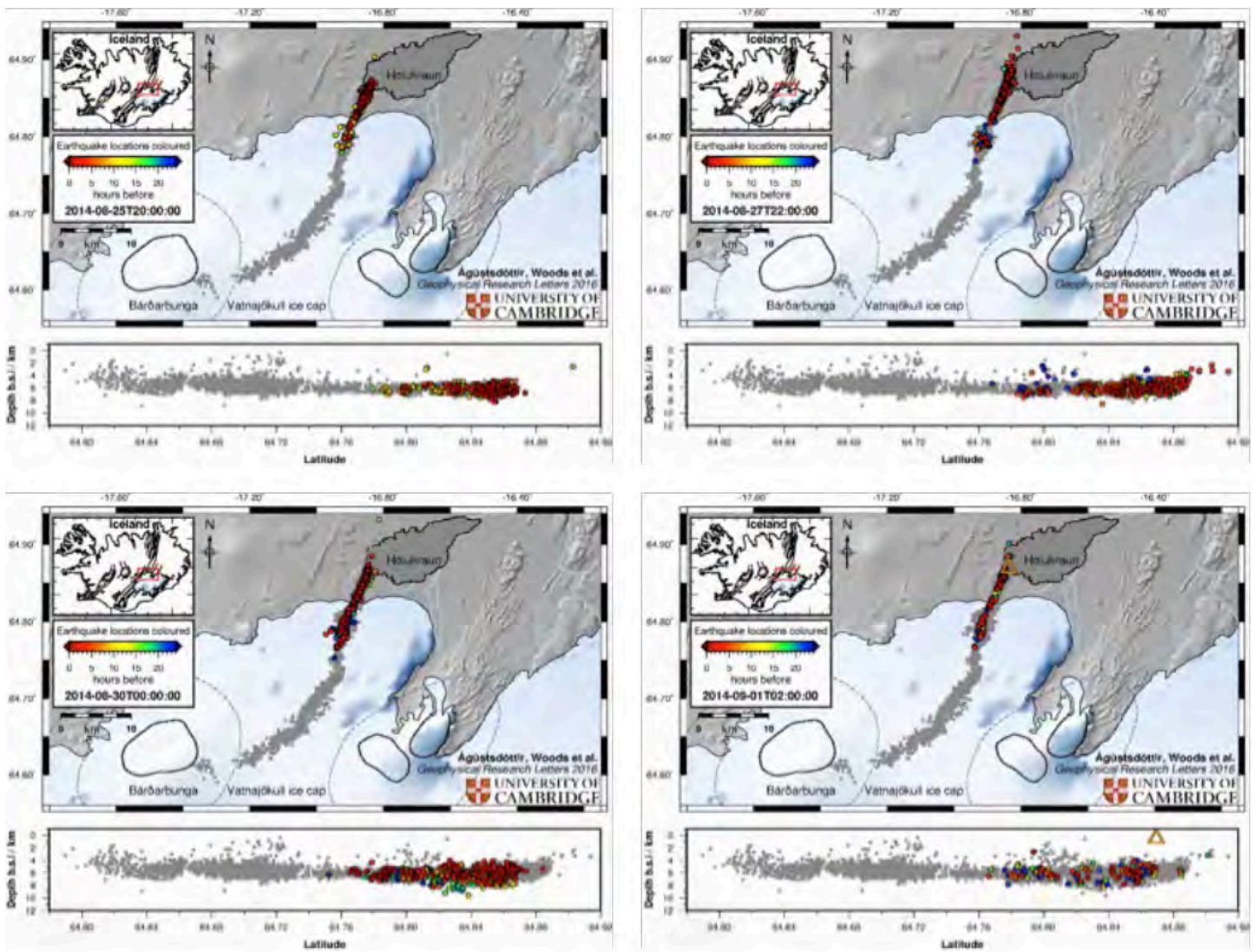


Figure 3: Time lapse images of earthquake locations shows the progression of the magma underground

to be continued... in lab!

5

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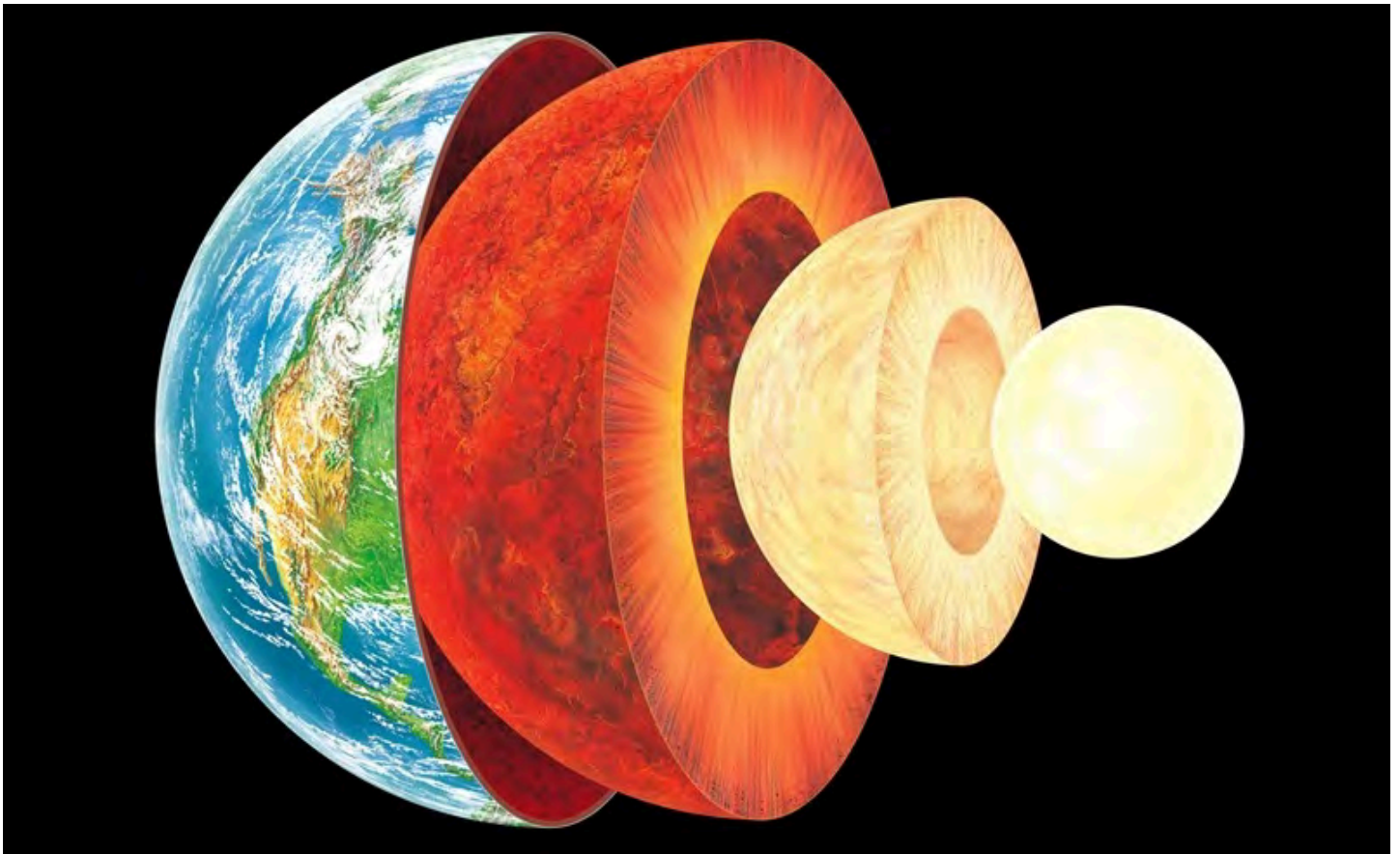
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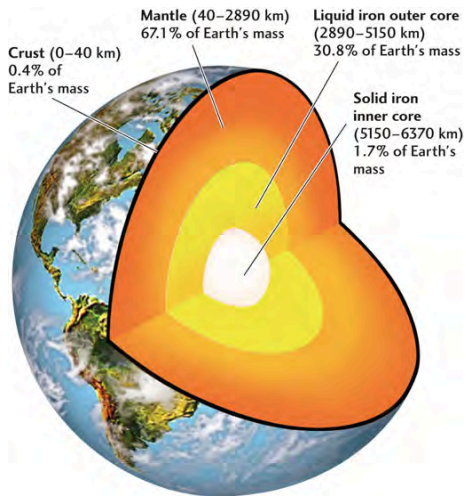
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G4P3.1

The general composition of internal Earth

We saw in the previous chapter ([G3.4 \(https://catcourses.ucmerced.edu/courses/32116/pages/g3-dot-4\)](https://catcourses.ucmerced.edu/courses/32116/pages/g3-dot-4)) that the internal Earth is made up of three different chemical layers: the crust, the mantle, and the core. The following artistic figure (*Credit: discovermagazine.com*) represents each of these layers (it also differentiates the inner core from the outer core):





- Core: it is divided in two parts: the solid, inner core and the liquid, outer core. Both inner and outer core are essentially composed of iron.
- Mantle: this chemical layer contains mainly oxygen, silicon, and magnesium.
- Crust: the top-surface layer of Earth is mostly composed of oxygen and silicon.

*Understanding Earth,
Grotzinger, 7th Ed.*

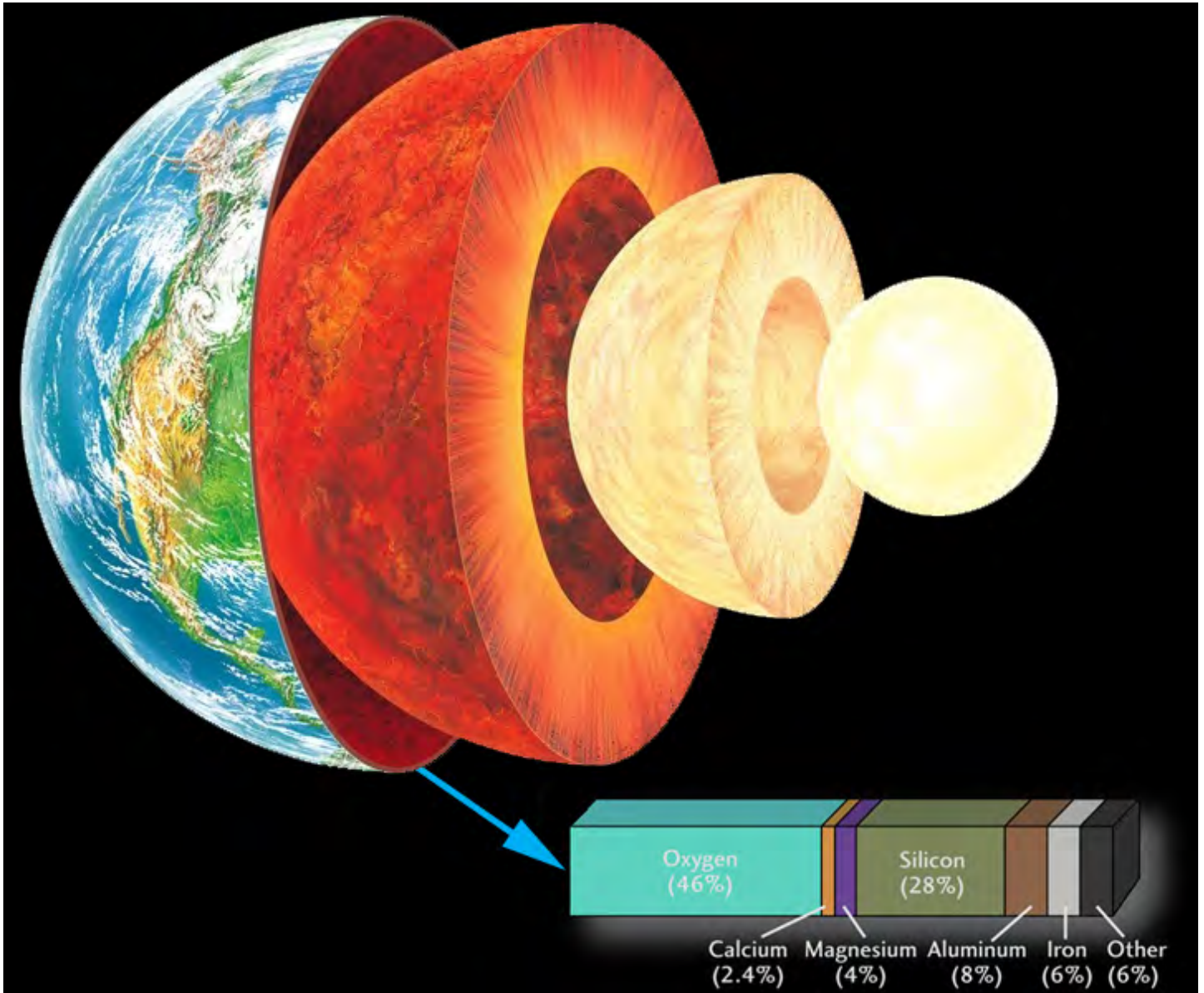
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G4P3.2

More details on the composition of the crust



Credits: *discovermagazine.com* & *Understanding Earth*, Grotzinger, 7th Ed.

Oxygen and silicon are the major elements of the composition of the crusts, making about 70% of continental crust by weight. Ten other elements composed the remaining 30%, meaning that the crust is composed of 12 different elements.

These 12 elements— oxygen, silicon, aluminum, iron, calcium, magnesium, sodium, potassium, titanium, hydrogen, manganese, and phosphorus—are called the **abundant elements** , and together they comprise 99.23% of the continental crust (cf. Table below). All of the remaining elements are referred to as **scarce elements**.

The 12 abundant elements are largely responsible for determining the composition of the common materials of Earth. For example, more than 4000 different minerals are known, but Earth's crust is overwhelmingly dominated by just 40 of them, most of which contain one or more of the 12 abundant elements.

TABLE 3.1 The Most Abundant Chemical Elements in the Continental Crust

Element	Ion	Percent by Weight
Oxygen (O)	O ²⁻	45.20
Silicon (Si)	Si ⁴⁺	27.20
Aluminum (Al)	Al ³⁺	8.00
Iron (Fe)	Fe ²⁺ and Fe ³⁺	5.80
Calcium (Ca)	Ca ²⁺	5.06
Magnesium (Mg)	Mg ²⁺	2.77
Sodium (Na)	Na ⁺	2.32
Potassium (K)	K ⁺	1.68
Titanium (Ti)	Ti ⁴⁺	0.86
Hydrogen (H)	H ⁺	0.14
Manganese (Mn)	Mn ²⁺ and Mn ⁴⁺	0.10
Phosphorus (P)	P ³⁺	0.10
All other elements		0.77
TOTAL		100.00

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G4P3.3

Minerals

Minerals are the building blocks of rocks of which the geosphere is made of.

Watch the following video that will introduce you to the definition of a mineral. It will also be mentioned of the mineral composition, and of the crystal structure that make a mineral unique. When watching the video, be sure to make some pause every time you need, and also take some notes!

[What is a mineral](https://www.youtube.com/watch?v=8a7p1NFn64s&t=1s) 

(<https://www.youtube.com/watch?v=8a7p1NFn64s&t=1s>)



(<https://www.youtube.com/watch?v=8a7p1NFn64s&t=1s>)

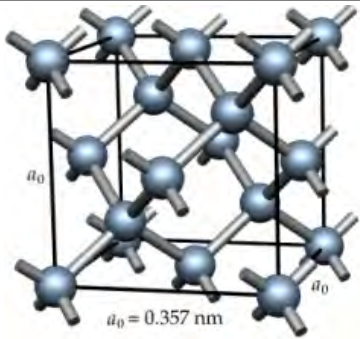
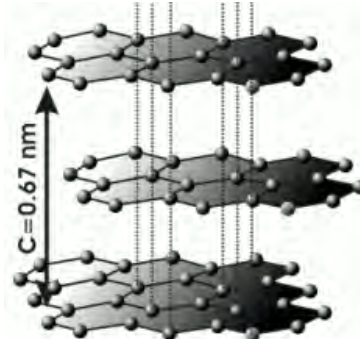
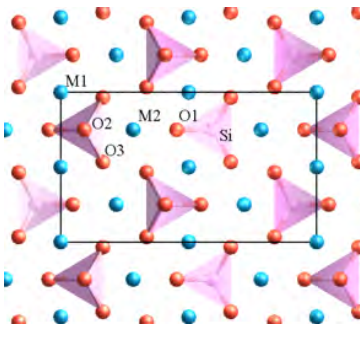
A substance is called a mineral when the five following requirements are met:

1. It must be a solid under normal Earth conditions
2. It must be naturally formed, not man-made (plastic is not a mineral).
3. It must be inorganic.
4. It must have a specific chemical composition, meaning that minerals are made up of chemical elements.
5. It must have a characteristic crystal structure, that is, an orderly internal arrangement of atoms.

Minerals are classified based on two important characteristics:

- Composition, which shows the different chemical elements that make the mineral, and the proportion of each element in the mineral;
- Crystal structure, which the way the atoms of the elements are packed together in the mineral.

Mineral	Picture	Composition	Crystal structure	Notes
<i>Example of two minerals with same chemical composition but different crystal structure:</i>				

Diamond		C (carbon atom)		Diamond is a cube with an octahedral cleavage. It has strong chemical bonds between all carbon atoms, ensuring rigid and perfect crystals.
Graphite		C (carbon atom)		The graphite consists of hexagonal sheaths which are parallel and do not have chemical bonds between them. As a result the graphite is soft, brittle and easily breakable.
<i>Example of two minerals with different chemical composition but same crystal structure:</i>				
Forsterite		Mg ₂ SiO ₄		Forsterite and Fayalite are two minerals from the Olivine family. Olivine is a series of minerals composed of iron-magnesium silicates ((Mg,Fe) ₂ SiO ₄).
Fayalite		Fe ₂ SiO ₄	Oxygen is shown in red, silicon in pink, magnesium or iron in blue.	The ratio of magnesium to iron varies between the two endmembers of the series: Forsterite (Fo) (magnesium silicate) and fayalite (Fa) (iron silicate).

Pictures from mindat.org

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Hi, everyone in this video. We're going to be talking about Minerals, so let's get started
What are minerals anyway well in the simplest sense of the term minerals are the building blocks of rocks?

Here's a rock. This is a piece of the igneous rock called granite if you look carefully at this chunk of granite

You'll see all sorts of speckles of Gray and white and black and if you [zoom] in

You'll see that these are [actually] crystals mineral crystals this particular piece [of] granite has the mineral quartz

Biotite Mica and Plagioclase feldspar

We'll study these more detail later on

so on Earth [there] are over 2,000 varieties of minerals some of which are seen here as

You can tell they come in a wide variety of colors shapes sizes

textures and all Sorts of different physical characteristics

Minerals have quite a few uses in our everyday life just to give you an example this here is an rough uncut diamond of

Course we use diamonds for jewelry we also use diamonds in construction [applications] as the tips of saw blades

And variety other industrial uses

Here's another mineral. This is called fluorite

most famous for its inclusion in

Toothpaste yes, when you see the name fluoride on a toothpaste that means it's made out of the mineral fluorite

But what else are minerals used for well this here is talc?

Talc is a very common mineral that is ground into a fine powder. It's very soft, and it's used in baby powder

Sometimes you might hear it referred to as talcum powder

One more example. This is a mineral selenite gypsum

Which is used all around us in drywall. This is the material that's used to build the walls inside Modern houses and buildings

Criteria for Minerals

But what exactly makes a mineral a mineral

Well to be considered a mineral the substance must meet five criteria

Let's go through those five [criteria] right now

number one

Substance must exist as a solid under normal conditions on Earth this means if you have [a] liquid or gas

Under normal conditions it cannot be considered a mineral it must be in the solid state

number two the substance must be

Naturally occurring on Earth this means it cannot be man-made

So for example plastic is not considered a mineral because it doesn't exist naturally. It's created by humans

Must be naturally-occurring

number three the substance must be in

organic meaning not coming from living or made of living things

so if you talk about

I don't know tree branches or leaves

They cannot be considered minerals because they are organic that is they came from living organisms

another example would be coral Coral is made by [small] sea creatures and

Therefore its organic material and cannot be considered a mineral minerals must be inorganic

number four the substance must have a

Fixed chemical formula meaning it's made up of a specific combination of elements

Let me give you an example the mineral quartz is composed of silicon and Oxygen bonded together

Specifically one silicon bonded to two oxygen atoms all quartz is made of this chemical formula

another example pyrite

Often known as fool's gold has a chemical formula of FeS_2 which is iron and s which [is] sulfur and when these are bonded together in this particular arrangement you get pyrite

so criteria for is that the substance must have a

Specific or fixed chemical formula has to be made of a specific recipe if you will of elements

And then finally criteria number five the atoms that make up the substance must be arranged in an orderly

Crystal structure a specific structure let me give you an example of this

[the] majority [of] the Minerals that [exist] on Earth's surface are considered silicates

And that means they're made up of in part silicon and oxygen

the Silica Tetrahedra is the most common arrangement of

Silicon and oxygen atoms within a mineral it looks something like this model here where the red balls of clay?

Represent Oxygen atoms and the Gray is a silicon atom and you can see they're bound together in this

Tetrahedral shape and this becomes the most common building block of Minerals on Earth

So that's what we mean when we say the atoms must be made up of a specific orderly structure

I'll give you another [example] of that in [just] a minute

So let's test ourselves a little bit here are those five criteria must be solid

Naturally occurring Inorganic have a fixed chemical formula and a specific atomic arrangement

So what about this pool of liquid Mercury can this be considered a mineral?

Well if you look at Criterion number one it says it must be a solid and this clearly is a liquid so

Sorry mercury. You're not a mineral

What about this chunk of bituminous coal?

Well, if you know anything about coal, you would know that it's actually formed from ancient

Tropical plants that have been compressed and squeezed together for millions of years

so it is [a] solid and it's certainly naturally occurring but
Because it's made from plants it is organic and therefore. It doesn't meet criteria number three and so coal is not
Considered a mineral
Well, what about ice this is an interesting one. Let's go through our Criteria. Is ice solid yes? And it does exist as a solid [on] at least some parts of the Earth the poles
Specifically and high up at mountains you get water that exists naturally as a solid so that's fine
Is it naturally occurring of course?
There's quite a bit of ice naturally existing on Earth it is not an organic material
It is not living it never was living and it's not made by living things
It does have a fixed chemical formula H_2O and it does have a specific atomic arrangement
So is it a mineral well according to our criteria? Yes, it is
However, there's a lot of debate about this and some people think ice should not be considered it because most places on Earth it would
Exist [as] [a] liquid [so] we'll give that one a question mark
What about this substance this is sulfur as you can see sulfur exists as a solid. That's good
It's naturally occurring it forms along volcanoes [it] is Inorganic
It's not living and it never was living it has a specific chemical formula
It's composed of the element sulfur, and it has a very specific
Atomic arrangement if we could zoom in we would see the atoms arranged in a specific way
So for our criteria to be met sulfur works, so it is considered a mineral

Physical Properties

now let's move on I
Want you to keep in mind that all [of] the physical properties of a mineral and what I mean by that is the colors
the shapes the textures the smells and the tastes the
appearance the Hardness the [sheen] all of these physical
characteristics of minerals
Result from one specific [thing] and that is the internal
arrangement of the atoms
To give you an example that sulfur
We were just looking at is yellow the reason it appears yellow is because of how these sulfur atoms are arranged inside
Quartz sometimes appear clear, and that's because of how the atoms are arranged the mineral halite
Tastes salty because of how the atoms are arranged
The Mineral sulfur again has kind of a rotten egg smell and that's a result of the internal arrangement of atoms
So all of the physical properties result from how those atoms are arranged. Let me give you one really neat example of this
This is a diamond the hardest mineral that exists on Earth
Fairly Rare in

Nice complete crystals it is the hardest substance

It has a hardness of ten on something called the mohs hardness scale which we will learn about

now

Diamond is composed of one element and that is carbon if you could zoom down inside this diamond

You would see the carbon atoms arranged in a pattern like this

Notice how all the atoms are interlocking connected to one another [this] creates a really strong bond

Which is what makes it in such a hard mineral?

But let's look at a different mineral

Did you ever wonder what made a pencil right a lot of people have the misconception that [it's] lead?

It's actually not lead it's a mineral known as graphite

Interestingly enough graphite like diamond is made up of only carbon

But if you look at how the carbon atoms are arranged, they're arranged into these sheets which are not very well connected

The result is that though it's made of the same elements as diamond. It's a much much softer

weaker Mineral

So again what gives these minerals their physical characteristics is how the atoms are arranged inside?

So let's do a quick recap

First thing we talked about was [how] minerals are the building blocks of rocks how they have lots and lots of uses on Earth

To be considered a mineral it must meet our five criteria

It must be a solid it must be naturally-occurring it must be Inorganic

It must have a specific composition and finally it must have a definite structure

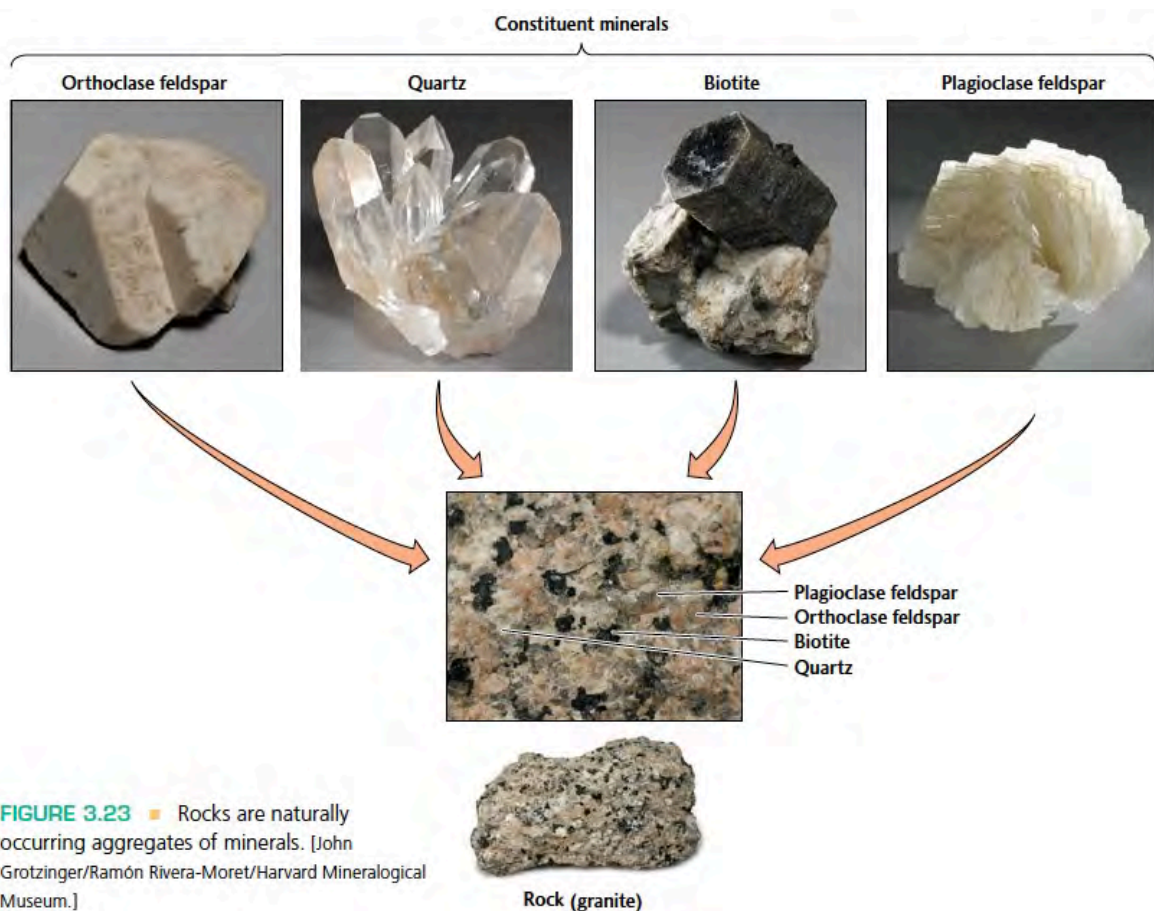
Thanks for listening

G4P3.4

Rocks

What is a rock?

If you look carefully at a rock such as the one shown below, you will certainly see that this rock formed by the aggregation of minerals. These minerals are joined in such a way that they retain their individual identity.



In some cases, rocks are not formed by the cementation of minerals, but by non mineral solid matter. These rocks are typically glassy volcanic rocks, called obsidian and pumice.

A more specific case is coal, a rock made only of compacted plant remains.



Obsidian, [geology.com](https://www.geology.com)



Pumice, [geology.com](https://www.geology.com)



Coal, [geology.com](https://www.geology.com)

What determines the physical appearance of a rock?

Rocks vary in color, in the sizes of their crystals or grains, and in the kinds of minerals that compose them. Therefore, the identity of a rock is determined partly by its mineralogy and partly by its texture:

- **Mineralogy** refers to the relative proportions of a rock's constituent minerals;
- **Texture** describes the sizes and shapes of a rock's mineral crystals or grains and the way they are put together.

The three families of rocks

The mineralogy and texture that determine a rock's appearance are themselves determined by the rock's geologic origin: where and how it formed (cf. Figure below). Three families of rocks are defined based on three distinct geologic environments and geologic processes:

Igneous rocks:

Igneous rocks are formed by the cooling and consolidation of magma. Magma is molten rock; it is one of the important liquid Earth materials. Magma forms deep underground. When it reaches the surface and pours out through a vent or crack, we refer to the molten rock as lava, and the vent through which it emerges is called a volcano.

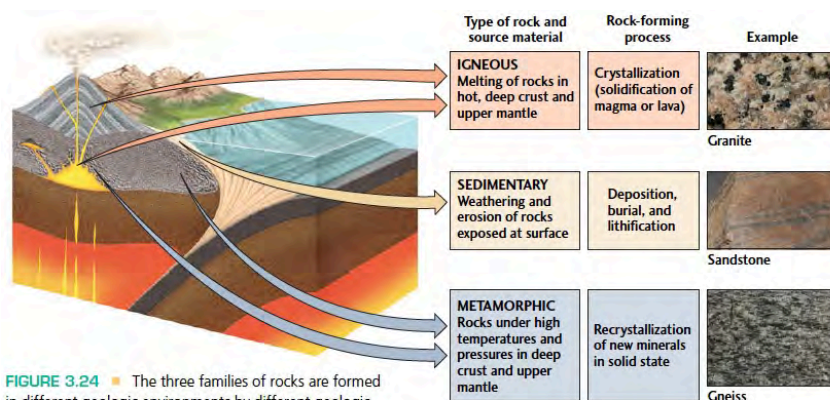


FIGURE 3.24 The three families of rocks are formed in different geologic environments by different geologic processes. [granite and gneiss: John Grotzinger/Ramón Rivera-Morel/ Harvard Mineralogical Museum; sandstone: John Grotzinger/Ramón Rivera-Morel/MIT.]

Sedimentary rocks:

Sedimentary rocks are formed either by chemical precipitation of material carried in solution in sea, lake, or river water, or by the deposition of mineral particles transported in suspension by water, wind, or ice.

Metamorphic rocks:

Metamorphic rocks are either igneous or sedimentary rocks that have been changed as a result of high temperatures, high pressures, or both.

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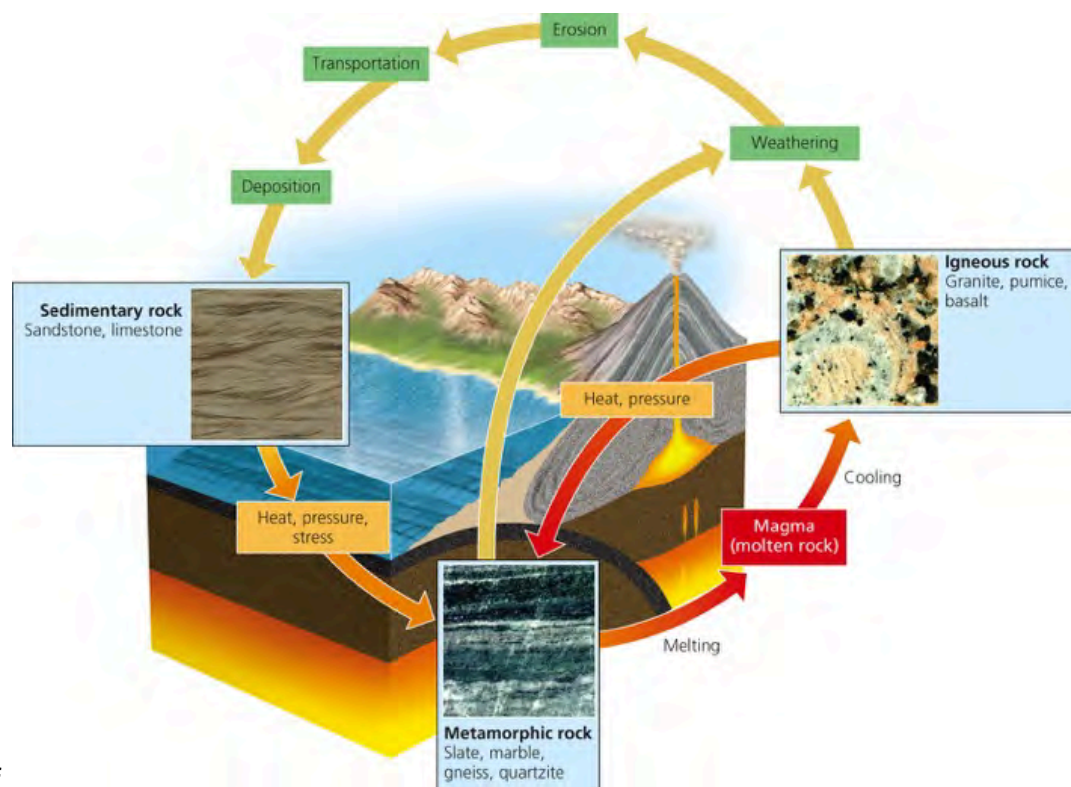
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G3P3.5

The rock cycle summarizes it all

Earth scientists have known for over 200 years that the three families of rocks—igneous, metamorphic, and sedimentary—all can evolve from one to another. Their observations gave rise to the concept of a rock cycle, which explains how each type of rock is transformed into one of the other two types. The rock cycle is now known to be the result of interactions between two of the three global geosystems: the plate tectonic system and the climate system.

Interactions between these two geosystems drive transfers of materials and energy among Earth's interior, the land surface, the ocean, and the atmosphere. For example, the formation of magmas at subduction zones results from processes



operating within the
plate tectonic system.

When these magmas erupt, materials and energy are transferred to the land surface, where the materials (newly formed rocks) are subject to weathering by the climate system. The eruption process also injects volcanic ash and carbon dioxide gas high into the atmosphere, where they may affect global climates. As global climates change, perhaps becoming warmer or cooler, the rate of weathering changes, which in turn influences the rate at which materials (sediments) are returned to Earth's interior.

The rock cycle never ends. It is always operating at different stages in various parts of the world, forming and eroding mountains in one place and laying down and burying sediments in another. The rocks that make up the solid Earth are recycled continuously, but we can see only the surface parts of the cycle. We must deduce the recycling of the deep crust and the mantle from indirect evidence.

This final video concludes with this chapter 3 and with all the Geosphere module. It will go over the material you covered over this chapter and the past ones, just in 4 minutes!

Thus, be sure to make some pause as needed, and take some good notes!

[Rocks and Minerals.mp4](https://catcourses.ucmerced.edu/courses/32116/files/7102763/download?wrap=1) (<https://catcourses.ucmerced.edu/courses/32116/files/7102763/download?wrap=1>)  (https://catcourses.ucmerced.edu/courses/32116/files/7102763/download?download_frd=1)

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